

Zeshui Song, 2/28/26, HW4

3.13 Under normal operating conditions, the electric motor exerts a torque of 2.8 kN·m on shaft AB. Knowing that each shaft is solid, determine the maximum shearing stress in (a) shaft AB, (b) shaft BC, (c) shaft CD. (Note: Torque from bearings at E and middle of shaft AB is very small and can be ignored.)

Given:

- $T_A = 2.8 \text{ kN} \cdot \text{m}$, $T_B = 1.4 \text{ kN} \cdot \text{m}$, $T_C = 0.9 \text{ kN} \cdot \text{m}$, $T_D = 0.5 \text{ kN} \cdot \text{m}$
- $d_{AB} = 56 \text{ mm}$, $d_{BC} = 48 \text{ mm}$, $d_{CD} = 48 \text{ mm}$, $d_{DE} = 46 \text{ mm}$
- Each shaft is solid

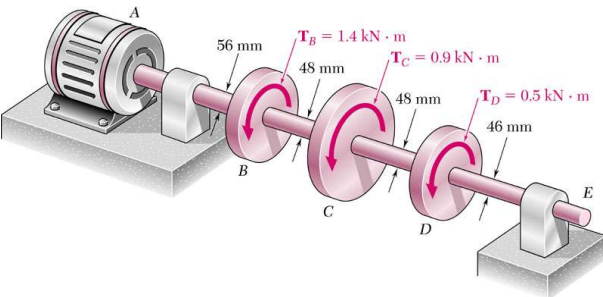
Find:

- Max shearing stress in AB, BC, CD

Approach:

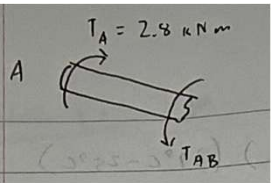
- Using statics, find the internal torques T_{AB} , T_{BC} , T_{CD}
- Apply the formula to find the max shearing stress

$$\tau_{\max} = \frac{c}{J} T$$



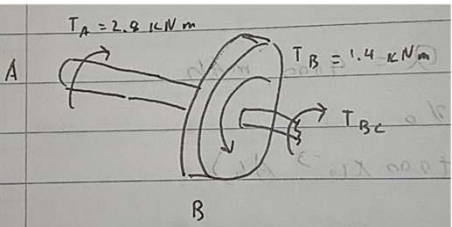
1

Zeshui Song, 2/28/26, HW4



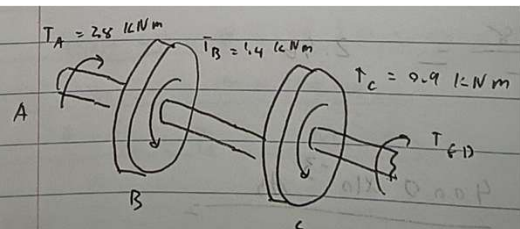
$$\sum T = 0 = T_{AB} - T_A$$

$$T_{AB} = 2.8 \text{ kN} \cdot \text{m} \text{ (CCW)}$$



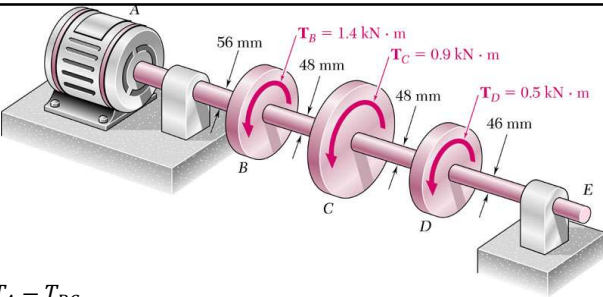
$$\sum T = 0 = T_B - T_A - T_{BC}$$

$$T_{BC} = 1.4 \text{ kN} \cdot \text{m} \text{ (CCW)}$$



$$\sum T = 0 = -T_A + T_B + T_C - T_{CD}$$

$$T_{CD} = 0.5 \text{ kN} \cdot \text{m} \text{ (CCW)}$$



2

Zeshui Song, 2/28/26, HW4

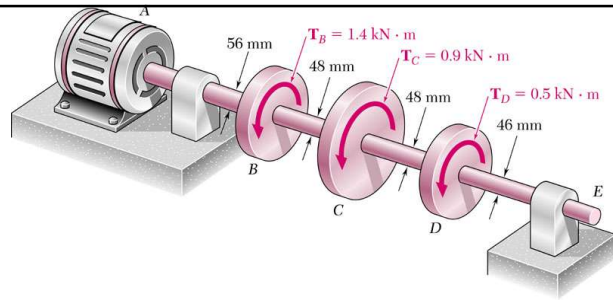
(a) $\tau_{AB,max}$

$$\tau_{max} = \frac{c}{J} T \quad J = \frac{\pi}{2} c^4$$

$$\tau_{AB,max} = \frac{2}{\pi \left(\frac{d_{AB}}{2}\right)^3} T_{AB} = \frac{16}{\pi (d_{AB})^3} T_{AB}$$

$$\tau_{AB,max} = \frac{16}{\pi \left[56 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}}\right]^3} [2.8 \text{ kN} \cdot \text{m}] \left[\frac{10^3 \text{ N}}{1 \text{ kN}}\right] = 81201501.58 \text{ Pa}$$

$$\tau_{AB,max} = 81.201 \text{ MPa}$$



3

Zeshui Song, 2/28/26, HW4

(b) $\tau_{BC,max}$

$$\tau_{BC,max} = \frac{16}{\pi (d_{BC})^3} T_{BC}$$

$$\tau_{BC,max} = \frac{16}{\pi \left[48 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}}\right]^3} [1.4 \text{ kN} \cdot \text{m}] \left[\frac{10^3 \text{ N}}{1 \text{ kN}}\right] = 64472488.52 \text{ Pa}$$

$$\tau_{BC,max} = 64.472 \text{ MPa}$$

(c) $\tau_{CD,max}$

$$\tau_{CD,max} = \frac{16}{\pi (d_{CD})^3} T_{CD}$$

$$\tau_{CD,max} = \frac{16}{\pi \left[48 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}}\right]^3} [0.5 \text{ kN} \cdot \text{m}] \left[\frac{10^3 \text{ N}}{1 \text{ kN}}\right] = 23025888.76 \text{ Pa}$$

$$\tau_{CD,max} = 23.025 \text{ MPa}$$

Since the max shear stress is at shaft AB, considering the stress concentration will be along the joint between the shaft and the disk at B, the critical location is most likely there.

[Time spent: 30 mins]

4

Zeshui Song, 2/28/26, HW4

3.25 Under normal operating conditions a motor exerts a torque of magnitude $T_F = 150 \text{ N}\cdot\text{m}$ at F . The shafts are made of a steel for which the allowable shearing stress is 75 MPa . Given the gear radii $r_D = 200 \text{ mm}$ and $r_G = 75 \text{ mm}$, determine the required diameter of (a) shaft CDE , (b) shaft FGH .

Given:

- $T_F = 150 \text{ N}\cdot\text{m}$
- $\tau_{all} = 75 \text{ MPa}$
- $r_D = 200 \text{ mm}$
- $r_G = 75 \text{ mm}$

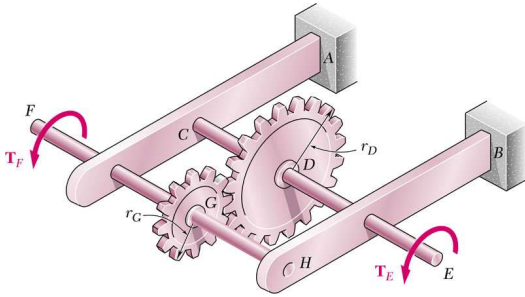
Find:

- Required diameter of shaft CDE and FGH

Approach:

- The torque at G will be equal to T_F
- Using gear ratios, I can find the applied torque on D from G
- Using statics I can find the internal torques T_{FGH} , T_{CDE}
- Setting max torque equal to allowed torque, I can find the diameters

$$\tau_{max} = \frac{c}{J} T$$

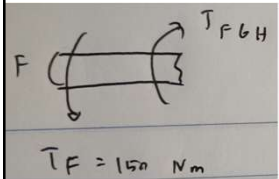


5

Zeshui Song, 2/28/26, HW4

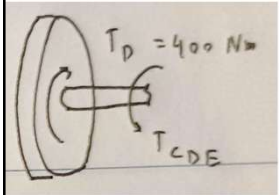
$$T_F = T_G = 150 \text{ N}\cdot\text{m}$$

$$T_D = \frac{r_D}{r_G} T_G = \frac{[200 \text{ mm}]}{[75 \text{ mm}]} [150 \text{ N}\cdot\text{m}] = 400 \text{ N}\cdot\text{m}$$



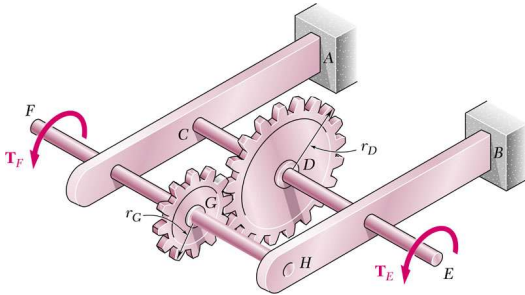
$$\sum T = 0 = T_F - T_{FGH}$$

$$T_{FGH} = 150 \text{ N}\cdot\text{m} \text{ (CW)}$$



$$\sum T = 0 = T_{CDE} - T_D$$

$$T_{CDE} = 400 \text{ N}\cdot\text{m} \text{ (CCW)}$$



6

Zeshui Song, 2/28/26, HW4

(a) d_{CDE}

$$\tau_{max} = \frac{c}{J}T = \frac{c}{\frac{\pi}{2}c^4}T = \frac{2}{\pi c^3}T = \frac{16}{\pi d^3}T$$

$$\tau_{CDE,max} = \tau_{all} = \frac{16}{\pi(d_{CDE})^3}T_{CDE}$$

$$d_{CDE} = \sqrt[3]{\frac{16T_{CDE}}{\tau_{all}\pi}} = \sqrt[3]{\frac{16[400 \text{ N} \cdot \text{m}]}{\pi[75 \text{ MPa}]\left[\frac{10^6 \text{ Pa}}{1 \text{ MPa}}\right]}} = 0.03006 \text{ m}$$

(a) d_{FGH}

$$d_{FGH} = \sqrt[3]{\frac{16T_{FGH}}{\tau_{all}\pi}} = \sqrt[3]{\frac{16[150 \text{ N} \cdot \text{m}]}{\pi[75 \text{ MPa}]\left[\frac{10^6 \text{ Pa}}{1 \text{ MPa}}\right]}} = 0.02167 \text{ m}$$

Since the internal tension is smaller for shaft FGH, it makes sense that it requires a smaller diameter to sustain an equal allowable shear stress.

[Time spent: 20 mins]

7

Zeshui Song, 2/28/26, HW4

3.36 The electric motor exerts a 500 N·m torque on the aluminum shaft *ABCD* when it is rotating at a constant speed. Knowing that $G = 27 \text{ GPa}$ and that the torques exerted on pulleys B and C are as shown, determine the angle of twist between (a) B and C, and (b) B and D.

Given:

- $T_D = 500 \text{ N} \cdot \text{m}$
- Shaft *ABCD* is rotating at constant speed (static equilibrium)
- $G = 27 \text{ GPa}$
- $L_{BC} = 1.2 \text{ m}$, $d_{BC} = 44 \text{ mm}$
- $L_{CD} = 0.9 \text{ m}$, $d_{CD} = 48 \text{ mm}$

Find:

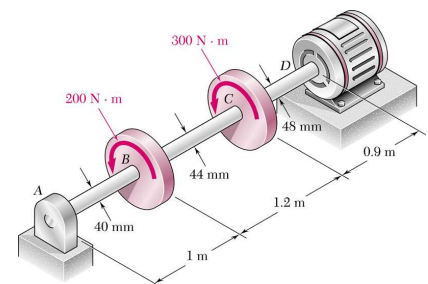
- Angle of twist ϕ_{BC} , ϕ_{BD}

Approach:

- Using statics to find internal torques T_{BC} , T_{CD}
- Using this equation to determine the angle of twist for BC and CD individually

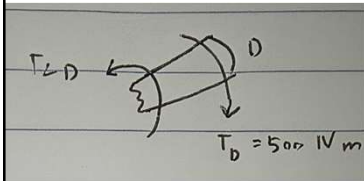
$$T = \frac{GJ\phi}{L}$$

- Add the angle of twist for BC and CD for angle of twist of BD



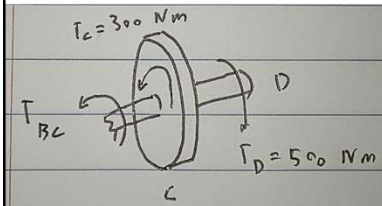
8

Zeshui Song, 2/28/26, HW4



$$\sum T = 0 = T_{CD} - T_D$$

$$T_{CD} = 500 \text{ N} \cdot \text{m} \text{ (CCW)}$$



$$\sum T = 0 = T_{BC} + T_C - T_D$$

$$T_{BC} = 200 \text{ N} \cdot \text{m} \text{ (CCW)}$$

9

Zeshui Song, 2/28/26, HW4

(a) ϕ_{BC}

$$T = \frac{GJ\phi}{L} = \frac{G \frac{\pi}{2} c^4 \phi}{L} = \frac{\pi}{2} G \left(\frac{d}{2}\right)^4 \phi = \frac{\pi}{32} G d^4 \phi \rightarrow \phi = \frac{32TL}{\pi G d^4}$$

$$\phi_{BC} = \frac{32T_{BC}L_{BC}}{\pi G d_{BC}^4} = \frac{32[200 \text{ N} \cdot \text{m}][1.2 \text{ m}]}{\pi \left[27 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}}\right] \left[44 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}}\right]^4} = 0.02415 \text{ rad (CCW)}$$

$$\boxed{\phi_{BC} = 1.383 \text{ deg}}$$

Although shaft BD is shorter and wider, it carries much more torque so that it has a larger twist angle than shaft BC.

[Time spent: 20 mins]

(b) ϕ_{BD}

$$\phi_{CD} = \frac{32T_{CD}L_{CD}}{\pi G d_{CD}^4} = \frac{32[500 \text{ N} \cdot \text{m}][0.9 \text{ m}]}{\pi \left[27 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}}\right] \left[48 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}}\right]^4} = 0.03198 \text{ rad (CCW)}$$

$$\phi_{CD} = 1.832 \text{ deg}$$

$$\boxed{\phi_{BD} = \phi_{BC} + \phi_{CD} = 3.215 \text{ deg}}$$

10

Zeshui Song, 2/28/26, HW4

3.69 One of two hollow drive shafts of a cruise ship is 40 m long, and its outer and inner diameters are 400 mm and 200 mm, respectively. The shaft is made of a steel for which $\tau_{all} = 60 \text{ MPa}$ and $G = 77.2 \text{ GPa}$. Knowing that the maximum speed of rotation of the shaft is 160 rpm, determine (a) the maximum power that can be transmitted by one shaft to its propeller, (b) the corresponding angle of twist of the shaft.

Given:

- $L = 40 \text{ m}$, $d_i = 200 \text{ mm}$, $d_o = 400 \text{ mm}$
- $\tau_{all} = 60 \text{ MPa}$
- $G = 77.2 \text{ GPa}$
- $\omega_{max} = 160 \text{ rpm}$

Find:

- Max power transmitted
- Twist angle at max power

Approach:

- Power transmitted by a shaft is

$$\dot{W} = T\omega$$

- Thus I can find the torque using τ_{all} from

$$\tau_{max} = \frac{c}{J} T$$

- This torque can also be used to find the twist angle from

$$T = \frac{GJ\phi}{L}$$

11

Zeshui Song, 2/28/26, HW4

$$J = \frac{\pi}{32} (d_o^4 - d_i^4) = \frac{\pi}{32} \left(\left[400 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right]^4 - \left[200 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right]^4 \right) = 0.002356194 \text{ m}^4$$

$$\tau_{max} = \tau_{all} = \frac{c}{J} T = \frac{\frac{d_o}{2}}{\frac{\pi}{32} (d_o^4 - d_i^4)} T \rightarrow T = \frac{\pi \tau_{all} (d_o^4 - d_i^4)}{16 d_o}$$

$$T = \frac{\pi \left[60 \text{ MPa} \cdot \frac{10^6 \text{ Pa}}{1 \text{ MPa}} \right] \left(\left[400 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right]^4 - \left[200 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right]^4 \right)}{16 \left[400 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right]} = 706858.347 \text{ N} \cdot \text{m}$$

(a) Max power

$$\dot{W} = T\omega = [706858.347 \text{ N} \cdot \text{m}] \left[160 \frac{\text{rot}}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ sec}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rot}} \right] = 11843525.28 \text{ W}$$

$$\boxed{\dot{W} = 11.843 \text{ MW}}$$

12

Zeshui Song, 2/28/26, HW4

(b) Angle of twist

$$T = \frac{GJ\phi}{L} \rightarrow \phi = \frac{TL}{GJ}$$

$$\phi = \frac{TL}{GJ} = \frac{[706858.347 \text{ N} \cdot \text{m}][40 \text{ m}]}{\left[77.2 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}}\right][0.002356194 \text{ m}^4]} = 0.15544 \text{ rad}$$

$$\phi = 8.906 \text{ deg}$$

While 8 degrees of twist angle seems like a lot, this makes sense since the length of the shaft is 40 m long, which makes it more compliant to shear strain.

[Time spent: 20 mins]

13

Zeshui Song, 2/28/26, HW4

Problem 3.77 will be graded.

3.77 The two solid shafts and gears shown are used to transmit 16 hp from the motor at A operating at a speed of 1260 rpm, to a machine tool at D. Knowing that the maximum allowable shearing stress is 8 ksi, determine the required diameter (a) of shaft AB, (b) of shaft CD.

Given:

- $\dot{W} = 16 \text{ hp}$
- $\omega = 1260 \text{ rpm}$
- $\tau_{all} = 8 \text{ ksi}$

Find:

- Required diameter of shaft AB and CD

Approach:

- Using the equation for power transmitted by a shaft, I can find the torque within shaft AB

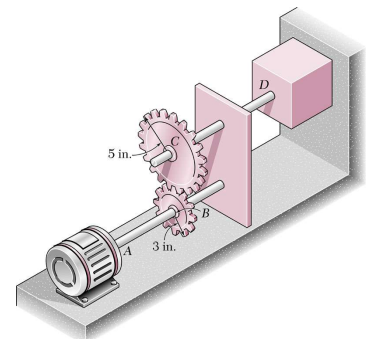
$$\dot{W} = T\omega$$

- I can use gear ratios at BC to determine the torque within shaft CD

$$T_{CD} = \frac{r_C}{r_B} T_{AB}$$

- Knowing the torques, I can solve for the diameters using the max shear stress

$$\tau_{max} = \frac{c}{J} T = \frac{c}{\frac{\pi}{2} c^4} T = \frac{2}{\pi c^3} T = \frac{16}{\pi d^3} T$$



14

Zeshui Song, 2/28/26, HW4

$$T_{AB} = \frac{\dot{W}}{\omega} = \frac{\left[16 \text{ hp} \cdot \frac{550 \text{ lbf} \cdot \text{ft/s}}{1 \text{ hp}}\right]}{\left[1260 \frac{\text{rot}}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rot}}\right]} = 66.6934 \text{ lbf} \cdot \text{ft}$$

$$T_{CD} = \frac{r_C}{r_B} T_{AB} = \frac{[5 \text{ in}]}{[3 \text{ in}]} [66.6934 \text{ lbf} \cdot \text{ft}] = 111.1558 \text{ lbf} \cdot \text{ft}$$

(a) Diameter of AB

$$\tau_{max} = \tau_{all} = \frac{16}{\pi d_{AB}^3} T_{AB}$$

$$d_{AB} = \sqrt[3]{\frac{16 T_{AB}}{\pi \tau_{all}}} = \sqrt[3]{\frac{16 \left[66.6934 \text{ lbf} \cdot \text{ft} \cdot \frac{12 \text{ in}}{1 \text{ ft}}\right]}{\pi \left[8 \text{ ksi} \cdot \frac{10^3 \text{ lbf/in}^2}{1 \text{ ksi}}\right]}}$$

$$d_{AB} = 0.798 \text{ in}$$

(b) Diameter of CD

$$d_{CD} = \sqrt[3]{\frac{16 T_{CD}}{\pi \tau_{all}}} = \sqrt[3]{\frac{16 \left[111.1558 \text{ lbf} \cdot \text{ft} \cdot \frac{12 \text{ in}}{1 \text{ ft}}\right]}{\pi \left[8 \text{ ksi} \cdot \frac{10^3 \text{ lbf/in}^2}{1 \text{ ksi}}\right]}}$$

$$d_{CD} = 0.946 \text{ in}$$

Due to the gear ratio at BC, it makes sense that there will be higher shear stress inside of shaft CD, thus requiring a slightly larger diameter. However, since there will be a stress concentration at the joint between the shafts and the gears, there may be a critical point along the joint on shaft CD.
[Time spent: 20 mins]

Note: writing is somewhat inaccurate; you are designing based on the same allowable shear stress in both shafts, so you shouldn't say there is a higher shear stress in CD. I know what you mean, but you could confuse someone.

15

Zeshui Song, 2/28/26, HW4

3.129 Determine the largest allowable square cross section of a steel shaft of length 6 m if the maximum shearing stress is not to exceed 120 MPa when the shaft is twisted through one complete revolution. Use a 77.2 GPa shear modulus.

Approach:

- Using the twist angle equation to find T:

$$\phi = \frac{TL}{c_2 a b^3 G}$$

- Using max shear stress equation for the square cross sectional area

$$\tau_{max} = \frac{T}{c_1 a b^2}$$

- Use table 3.1 for values of c_1, c_2

$$\frac{a}{b} = 1$$

- $c_1 = 0.208, c_2 = 0.1406$

Given:

- $L = 6 \text{ m}$
- $\tau_{all} = 120 \text{ MPa}$ (after 1 revolution of twist)
- $G = 77.2 \text{ GPa}$

Find:

- Largest allowable square cross section

TABLE 3.1. Coefficients for Rectangular Bars in Torsion

a/b	c_1	c_2
1.0	0.208	0.1406
1.2	0.219	0.1661
1.5	0.231	0.1958
2.0	0.246	0.229
2.5	0.258	0.249
3.0	0.267	0.263
4.0	0.282	0.281
5.0	0.291	0.291
10.0	0.312	0.312
∞	0.333	0.333

16

Zeshui Song, 2/28/26, HW4

$$\phi = \frac{TL}{c_2 ab^3 G} \rightarrow T = \frac{\phi c_2 ab^3 G}{L}, a = b$$

$$T = \frac{[2\pi \text{ rad}][0.1406]a^4 \left[77.2 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}} \right]}{[6 \text{ m}]} = 1.13666 \times 10^{10} a^4 \text{ N} \cdot \text{m}$$

$$\tau_{max} = \tau_{all} = \frac{T}{c_1 ab^2}, a = b \quad \tau_{all} = \frac{[1.13666 \times 10^{10} a^4 \text{ N} \cdot \text{m}]}{c_1 a^3}$$

$$a = \frac{\tau_{all} c_1}{1.13666 \times 10^{10}} = \frac{\left[120 \text{ MPa} \cdot \frac{10^6 \text{ Pa}}{1 \text{ MPa}} \right] [0.208]}{1.13666 \times 10^{10}} = 0.00219 \text{ m}$$

$$\boxed{a = 2.19 \text{ mm}}$$

Looking at the ratios of a/b, if the cross section becomes more rectangular, c_1, c_2 will increase, causing both the twist angle and max shear stress to increase. Thus, the square cross section is most efficient at resisting torques (outside of circular cross sections).

[Time spent: 25 mins]

17

Zeshui Song, 2/28/26, HW4

3.137+ A 750 N-m torque is applied to a hollow shaft having the ovoid cross section shown at right and a uniform 6 mm wall thickness. Neglecting the effect of stress concentrations, determine the shearing stress at points *a* and *b*.

+ If the shaft is constructed from 2014-T6 aluminum alloy and has a length of 0.8 m, estimate the twist angle. In your discussion of the results, compare the torsional stiffness to the torsional stiffness of a 30mm ID hollow round shaft plus the stiffnesses of two 60mm wide 6mm deep rectangular shafts.

Approach:

- Using the equation for shear stress in thin walled hollow shafts

$$\tau = \frac{T}{2ta}$$

- Using the twist angle equation

$$\phi = \frac{TL}{4Ga^2} \oint \frac{ds}{t}$$

- Comparing the torsional stiffness using

$$k_T = \frac{T}{\phi} = \frac{4Ga^2}{L \oint \frac{ds}{t}}$$

And for the round + rectangular shafts, use the same relation

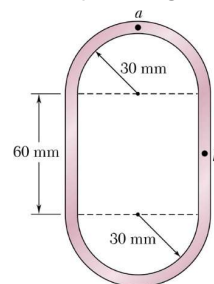
$$k_T = \frac{T}{\phi}$$

Given:

- Thickness $t = 6 \text{ mm}$
- $T = 750 \text{ N} \cdot \text{m}$
- 2014-T6 aluminum alloy ($G = 28 \text{ GPa}$)
- $L = 0.8 \text{ m}$

Find:

- Shear stress at *a* and *b*
- Estimate the twist angle
- Compare the torsional stiffness of the hollow shaft to the torsional stiffness of a 30 mm ID round shaft plus two 60mm wide, 6 mm deep rectangular shafts



18

Zeshui Song, 2/28/26, HW4

(a) Shear stress at a and b, where a is the area bounded by the *center line*

$$a = \left[\pi \left(33 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right)^2 + \left(66 \cdot 60 \text{ mm}^2 \cdot \frac{10^{-6} \text{ m}^2}{1 \text{ mm}^2} \right) \right] = 0.00738 \text{ m}^2$$

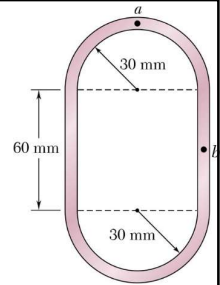
$$\tau = \frac{T}{2ta} = \frac{[750 \text{ N} \cdot \text{m}]}{2 \left[6 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right] [0.00738 \text{ m}^2]} = 8467464.29 \text{ Pa}$$

$$\tau = 8.467 \text{ MPa}$$

(b) Twist angle, where S is the perimeter of the *center line*

$$\phi = \frac{TL}{4Ga^2 t} = \frac{[750 \text{ N} \cdot \text{m}][0.8 \text{ m}]}{4 \left[28 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}} \right] [0.00738 \text{ m}^2]^2} \frac{\left[2\pi \left(33 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right) + 2 \left(60 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right) \right]}{\left[6 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right]} = 0.005364 \text{ rad}$$

$$\phi = 0.3073 \text{ deg}$$



19

Zeshui Song, 2/28/26, HW4

(c) Comparing stiffnesses:

Thin-walled hollow shaft:

$$k_T = \frac{T}{\phi} = \frac{[750 \text{ N} \cdot \text{m}]}{[0.005364 \text{ rad}]} = 139805.277 \text{ N} \cdot \text{m}$$

Round shaft:

$$k_T = \frac{T}{\phi} = \frac{GJ}{L} = \frac{G}{L} \frac{\pi}{32} (d_o^4 - d_i^4)$$

$$k_T = \frac{\left[28 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}} \right]}{[0.8 \text{ m}]} \frac{\pi}{32} \left(\left(36 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right)^4 - \left(30 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right)^4 \right) = 2988.1022 \text{ N} \cdot \text{m}$$

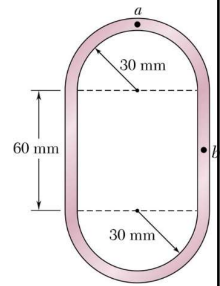
Two 60mm wide 6mm deep rectangular shafts. :

$$\frac{a}{b} = 10 \rightarrow c_2 = 0.312$$

$$k_T = 2 \frac{T}{\phi} = \frac{c_2 ab^3 G}{L} = 2 \frac{[0.312] \left[60 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right] \left[6 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right]^3 \left[28 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}} \right]}{[0.8 \text{ m}]} = 283.0464 \text{ N} \cdot \text{m}$$

Sum

$$k_T = 3271.148 \text{ N} \cdot \text{m}$$



20

Zeshui Song, 2/28/26, HW4

We see that the ovoid shaft is much stiffer than the sum of the round and rectangular shafts, this is mostly because the ovoid shaft is modeled as a continuous shaft instead of a collection of separate pieces. Especially since the rigidity of rectangular shafts are much weaker than round shafts.

[Time spent: 45 mins]

